

Project Planning – EduGenie-AI

Google Gemini Powered AI Learning Assistant

Page 1 – Planning Overview and Project Goals

The Project Planning phase defines how EduGenie-AI will be developed, tested, documented, and demonstrated. Planning converts the project design into an organized sequence of activities with clear deliverables, responsibilities, resources, and milestones.

Project Goals

- Develop a functional AI-powered learning assistant for students.
- Integrate Google Gemini with a FastAPI-based backend.
- Provide question answering, explanations, quizzes, summaries, and learning recommendations.
- Create a simple web interface using HTML, CSS, and JavaScript.
- Test the major features and handle common errors.
- Prepare complete project documentation and a final demonstration.

Planning Principles

- Divide the project into manageable phases.
- Complete and verify one major component before moving to the next.
- Keep source code modular so changes can be made safely.
- Track technical problems and resolve them during development.
- Maintain secure handling of API credentials.
- Reserve time for integration testing, documentation, and demonstration preparation.

Major Project Phases

Phase	Main Deliverable
Brainstorming & Ideation	Problem statement and proposed solution
Requirement Analysis	Functional and non-functional requirements
Project Design	Architecture, modules and interface design
Project Planning	Schedule, resources, risks and task allocation
Project Development	Working EduGenie-AI application
Project Testing	Test cases and verified functionality
Project Documentation	Reports, screenshots and technical details
Project Demonstration	Final presentation and live demonstration

Page 2 – Work Breakdown and Task Planning

A Work Breakdown Structure (WBS) divides the project into smaller tasks. This makes development easier to organize and helps ensure that important activities are not missed.

Work Breakdown Structure

Work Package	Activities
1. Project Setup	Create project folder, virtual environment, dependencies and Git configuration
2. Backend Development	Create FastAPI application, routes and module connections
3. AI Integration	Configure Gemini API, model settings and prompt handling
4. Feature Development	Implement Q&A, explanation, quiz, summary and learning path features
5. Frontend Development	Build HTML interface, styling and JavaScript request handling
6. Integration	Connect frontend, backend and Gemini modules
7. Testing	Perform functional, integration, error and usability testing
8. Documentation	Prepare report, screenshots, architecture and usage instructions
9. Demonstration	Prepare presentation, test scenarios and final project walkthrough

Task Dependencies

Some tasks depend on earlier activities. For example, backend routes should be available before complete frontend integration can be tested. Gemini configuration should be completed before AI-powered features can be verified. Testing should occur throughout development rather than only at the end.

Planned Development Sequence

- Set up Python environment and required packages.
- Configure environment variables for Gemini.
- Develop and verify individual backend modules.
- Create FastAPI routes for each feature.
- Develop the frontend interface.
- Connect frontend requests to backend endpoints.
- Test each feature independently.
- Perform complete system integration testing.
- Prepare documentation and demonstration materials.

Page 3 – Schedule, Milestones and Resources

The project schedule provides a practical timeline for completing the major activities. The exact duration can be adjusted according to academic deadlines and team availability.

Proposed Project Schedule

Week	Planned Activity	Milestone
Week 1	Brainstorming and problem identification	Problem and solution finalized
Week 2	Requirement analysis	Requirements documented
Week 3	Project design and architecture	Design finalized
Week 4	Project setup and backend foundation	FastAPI application running
Week 5	Gemini integration and Q&A	AI response feature working
Week 6	Explanation, quiz and summary modules	Core learning features implemented
Week 7	Frontend and full integration	Complete application connected
Week 8	Testing and debugging	Major defects resolved
Week 9	Documentation and demonstration preparation	Final project package ready

Required Resources

- Computer or laptop capable of running Python and VS Code.
- Python environment with project dependencies.
- FastAPI and Uvicorn for backend development.
- Google Gemini API and Google GenAI Python SDK.
- HTML, CSS and JavaScript for the frontend.
- Git and GitHub for source-code version control.
- Web browser for application testing.
- Documentation and presentation tools for final submission.

Project Milestones

Important milestones include successful FastAPI startup, successful Gemini API communication, completion of each learning module, successful frontend-backend integration, completion of testing, and preparation of the final demonstration.

Page 4 – Team Roles, Risk Management and Quality Planning

Project planning also identifies responsibilities and possible risks. Even for an individual academic project, defining roles helps organize the work clearly.

Suggested Roles

Role	Responsibility
Project Lead	Coordinate activities and track overall progress
Backend Developer	Develop FastAPI routes and application logic
AI Integration Developer	Configure Gemini and design prompts
Frontend Developer	Create interface and client-side functionality
Tester	Prepare test cases and verify system behavior
Documentation Lead	Prepare report, diagrams, screenshots and user instructions
Presenter	Prepare final demonstration and explain project workflow

Risk Management

Risk	Possible Impact	Planned Mitigation
Gemini API unavailable	AI features may fail	Handle API errors and test connectivity early
Invalid API key	AI requests fail	Store key in .env and verify configuration
Model/service limits	Temporary request failures	Handle errors and retry later when appropriate
Frontend-backend mismatch	Features do not work correctly	Test endpoints independently before integration
Unexpected AI output	Learning response may be unclear	Use clear prompts and validate important outputs
Development delay	Milestones may slip	Prioritize core features and track tasks regularly
Code defects	Application errors	Use modular testing and debugging

Quality Planning

Quality will be maintained through incremental development, code organization, endpoint testing, user-interface checks, error handling, and final integration testing. Each major feature should be verified before it is considered complete.

Page 5 – Project Tracking, Deliverables and Completion Plan

Project tracking ensures that planned activities are actually completed. Progress can be monitored using a simple task list containing the task name, status, testing result, and remarks.

Progress Tracking Format

Task	Status	Verification
Project setup	Completed / Pending	Application environment runs
Gemini configuration	Completed / Pending	Test prompt returns response
Q&A module	Completed / Pending	Question receives answer
Explanation module	Completed / Pending	Topic explanation generated
Quiz module	Completed / Pending	Quiz content generated
Summary module	Completed / Pending	Summary generated
Learning recommendations	Completed / Pending	Recommendations generated
Frontend integration	Completed / Pending	UI communicates with API
Testing	Completed / Pending	Test cases verified
Documentation	Completed / Pending	Final report prepared
Demonstration	Completed / Pending	Demo flow rehearsed

Final Deliverables

- Working EduGenie-AI application.
- Source code containing the FastAPI backend and learning modules.
- Frontend interface files.
- Requirements and dependency configuration.
- Test cases and testing results.
- Project documentation and diagrams.
- Presentation or demonstration materials.
- Git repository containing the project source code, with secret configuration excluded.

Completion Criteria

The project can be considered ready for final submission when the main learning features work, the frontend and backend communicate correctly, Gemini requests are handled successfully, common errors are managed, tests have been completed, documentation is prepared, and the demonstration flow has been rehearsed.

Conclusion

The Project Planning phase gives EduGenie-AI a structured development roadmap. It defines the work breakdown, schedule, resources, responsibilities, risks, quality checks, and final deliverables. This plan provides a practical foundation for moving from design into systematic development and testing.